Andrew S. **Rosen**

Assistant Professor, Princeton University
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Academic Appointments

Princeton University *Princeton, NJ*
Assistant Professor, Department of Chemical and Biological Engineering Aug. 2024 – Present
Associated Faculty, Princeton Materials Institute
Associated Faculty, Princeton Institute for Computational Science and Engineering
Associated Faculty, Center for Statistics and Machine Learning
Associated Faculty, Andlinger Center for Energy and the Environment
Associated Faculty, Computational Sciences Department, Princeton Plasma Physics Laboratory

Education and Training

University of California, Berkeley *Berkeley, CA*
Miller Research Fellow, Department of Materials Science and Engineering 2021 – 2024
Mentor: Prof. Kristin A. Persson

Northwestern University *Evanston, IL*
Ph.D. in Chemical Engineering 2016 – 2021
Advisors: Prof. Randall Q. Snurr, Prof. Justin M. Notestein

Tufts University *Medford, MA*
B.S. in Chemical Engineering, *summa cum laude* 2011 – 2015

Honors & Awards

AFOSR Young Investigator Program (YIP) Award, Department of Defense 2026
AIChE 35 Under 35 Award, American Institute of Chemical Engineers 2026
Commendation for Outstanding Teaching – CBE/MSE 423, Princeton University – SEAS 2026
Commendation for Outstanding Teaching – CBE 504, Princeton University – SEAS 2026
Breakthrough Energy Explorer Fellow, Breakthrough Energy Foundation 2025
Commendation for Outstanding Teaching – CBE 504, Princeton University – SEAS 2025
Royal Society of Chemistry Outstanding Reviewer, Digital Discovery 2023
Distinguished Young Scholars Seminar, University of Washington – Chemical Engineering 2022
Miller Research Fellowship, University of California, Berkeley 2021
Presidential Fellowship, Northwestern University – The Graduate School 2021
CAS Future Leader, American Chemical Society 2020
Distinguished Graduate Researcher Award, Northwestern University CBE 2020
Outstanding Research Mentor Award, International Institute for Nanotechnology 2020
CoMSEF Graduate Student Award – Honorable Mention, AIChE CoMSEF 2020
ACS Division of Inorganic Chemistry Travel Award, ACS Division of Inorganic Chemistry 2020
ACS Catalysis Division Travel Award, ACS Catalysis Division 2019
Ryan Fellowship, International Institute for Nanotechnology 2018
George Thodos Teaching Assistant Award – ChBE 404, Northwestern University CBE 2018
George Thodos Teaching Assistant Award – ChBE 212, Northwestern University CBE 2017

National Defense Science and Engineering Graduate Fellowship, U.S. Dept. of Defense 2017
Class of 1947 Victor Prather Prize, Tufts University 2015
Goldwater Scholarship, Barry Goldwater Scholarship Foundation 2014
Summer Research Scholarship, Tufts University 2014
National Undergraduate Fellowship, Princeton Plasma Physics Laboratory 2013
Outstanding Poster Presentation Award, 55th APS Plasma Physics Conference 2013
Tau Beta Pi, Tufts University 2013

Publications

*corresponding author | †equal contribution | ᵘundergrad mentee | Google Scholar | ORCID: 0000-0002-0141-7006
> 7000 citations via Google Scholar, *h*-index = 29

Submitted Manuscripts and Pre-Prints

51. Pymatgen: 15 Years of Community Growth, New Functionality, and Future Prospects
J. George†, M.K. Horton†, A.D. Kaplan†, S.R. Kavanagh†, J. Riebesell†, **A.S. Rosen**†, H. Yang†, F. Adams, M. Amachraa, O. Andriuc, O.A. Ashour, S. Bajaj, E. Banyas, L. Barroso-Luque, K. Berket, N. Bernstein, A. Bhardwaj, A.H.U. Bhatti, S.M. Blau, B. Bocklund, A. Bonkowski, D. Broberg, G. Brunin, K. Bystrom, C.A. Cardot, R. Chang, C. Chen, W. Chen, X. Chen, Y. Chiang, K. Choudhary, O.A. Cohen, M.D. Costa, S.T. Dacek III, M. de Jong, T. Demeyere, A.K. Dozier, A. Dunn, S. Dwaraknath, M.T. Dylla, K. Eimre, A.R. Epstein, C. Ertural, M. Esters, M.L. Evans, J. Falke, A. Fekete, B. Field, A. Fonari, N.C. Frey, D.W. Gaines II, A.M. Ganose, M. Giantomassi, R.E.A. Goodall, P.-F. Gostin, A. Gupta, V.-A. Ha, G. Hautier, N. Heinsdorf, M.F. Herbst, T. Hou, P. Huck, Y. Ikeda, A.J. Jackson, S. Jayaraman, F.R. Kearns, S. Kim, R.S. Kingsbury, A.M. Krajewski, Y. Kumagai, M.C. Kuner, W.T. Laderer, W. Li, X.-G. Li, Y.H. Liang, H. Liu, K. Liu, M. Liu, A. Lutfi, O. Marchand, M.J. McDermott, B.K. Medasani, F. Meng, K.D. Miller, H. Miranda, R.S. Mohanakrishnan, J.H. Montoya, G.C. Moore, B.J. Morgan, J.M. Munro, K. Muraoka, A.A. Naik, W. O'Leary, A. Onwuli, J. Park, D.F. Parsons, K. Pei, G. Petretto, T.A.R. Purcell, J. Qi, X. Qu, I. Radchenko, N.N. Rajput, F. Ricci, B.S. Rich, W.D. Richards, G.-M. Rignanese, X. Ruan, G. Salvato Vallverdu, J. Schmidt, J.-X. Shen, R. Shimogawa, K. Shinohara, A.K. Singh, E. Sivonxay, S. Smeets, T.E. Smidt, A. Sobolev, E.W.C. Spotte-Smith, C. Stihl, W. Sun, J.D. Sundberg, Y. Suzuki, L. Talirz, H. Tang, E. Taw, C. Tezak, F. Therrien, S.D. Thiel, S.B. Torrisi, M.E. Turiansky, K. Ueltzen, M.J. van Setten, J. Vieten, S. Vijay, N. Wagner, F. Wan, A. Wang, J. Wang, Z. Wang, L. Ward, T. Warford, D. Waroquiers, M. Wen, O. Weser, N. Winner, D. Winston, M. Wolloch, B.M. Wood, R. Woods-Robinson, Y. Xiao, Y. Xie, F. Yalcin, R.X. Yang, W. Ye, P. Zarabadi-Poor, J. Zeng, H. Zhao, K. Zheng, H. Zhu, Z. Zhu, N.E.R. Zimmermann, D. Zügner, K.A. Persson, A. Jain, S.P. Ong*

50. Democratizing Atomistic Simulation Workflows for the AI Era with the Quantum Accelerator
A.S. Rosen*, N. Goyal, B. Ayers, V. Bansal, J.H. Baratta, S.M. Blau, Y. Chiang, S. Choi, O.A. Cohen, B. Dallmann, T. Demeyere, W. Engler, Y.-W. Fang, I. Furrick, E. Huerta, H. Kim, H. Kondo, A. Kumar, J. Kwon, O. Mamun, C.B. Musgrave III, H. Oliaei, A. Sahaᵘ, D. Sarpa, B.X. Shi, Y. Shi, X. Wang, R.B. Wexler. arXiv:2609.33823

49. Towards a Metal–Organic Framework with Pore-Confined Electrons
J.H. Baratta, **A.S. Rosen***. arXiv:2609.13939

48. Layered Double Hydroxides with Interlayer Electrons Activate Strong Chemical Bonds
Y. Shi, **A.S. Rosen***. DOI: 10.26434/chemrxiv.15008487/v1

47. Agentic Laboratories of the Future: Towards World Models for Scientific Discovery
K. Fu, Y. Wu, L. Lyu, S. Li, S. Huang, S. Xu, J. Zheng, X. Liu, S. Liu, G. Barbone, Y. Liu, L. Fan, Y. Zhu, M. Davis, K. Jiang, S. Yang, X. Kuang, Y. Luo, F. Dominici, C. Curtis, X. Qiu, A. Abadie, C. Uhler, M. Tang, J.C. Wu, J. Li, J. Toettcher, J.L. Avalos, R. Rojansky, H. Zhao, J.P.A. Ioannidis, B. Rand, E. Lundberg, **A.S. Rosen**, Z. Liu, S. Kelley, T. Montine, C.V. Theodoris, K.S. Pollard, F. Li, X. Zhang, T. Peng, D. Basov, X. Qian, E. Xing, J. Leskovec, N. Shah, Z. Bao, P. Perona, S. Wu, C. Brangwynne, L. Cong*, M. Wang*. DOI: 10.20944/preprints202608.0213.v1

46. Designing High-Valent Copper-Oxo Complexes Through Ligand Engineering
K. Yao†, X. Wang†, S. Iyerᵘ, **A.S. Rosen***, R.Q. Snurr*

45. The Open Molecules 2025 (OMol25) Dataset, Evaluations, and Models
D.S. Levine[†], M. Shuaibi[†], E.W.C. Spotte-Smith, M.G. Taylor, M.R. Hasyim, K. Michel, I. Batatia, G. Csányi, M. Dzamba, P. Eastman, N.C. Frey, X. Fu, V. Gharakhanyan, A.S. Krishnapriyan, J.A. Rackers, S. Raja, A. Rizvi, **A.S. Rosen**, Z. Ulissi, S. Vargas, C.L. Zitnick*, S.M. Blau*, B.M. Wood*. arXiv:2505.08762

Published and In Press

44. Fine-Tuning a Universal Machine-Learned Interatomic Potential for Oxygen Plasma Interactions with WS_2
J. Kwon, **A.S. Rosen**, D.B. Graves*, *Journal of Vacuum Science & Technology A*, 44, 062601 (2026).
— Selected an Editor's Pick

43. Predicting the Thermodynamic Limits of Metal–Organic Framework Metastability
B. Dallmann, A. Saha[u], **A.S. Rosen***, *Journal of the American Chemical Society*, 148, 19487–19501 (2026).

42. Beyond Big Data in Quantum Chemistry
A.S. Rosen*, *Nature Chemical Engineering*, 3, 79 (2026).

41. Capturing the Complexities of Catalyst–Support Interactions with the Help of Machine Learning
A.S. Rosen*, *Angewandte Chemie International Edition*, 64, e202521310 (2025).

40. Multi-Gas Adsorption with Single-Site Cooperativity in a Metal–Organic Framework
K.M. Carsch, H.Z.H. Jiang, R.A. Klein, **A.S. Rosen**, P. Summerhill, J.L. Peltier, A.J. Huang, R.A. Murphy, M.N. Dods, H.A. Silva, Z. Hasanbasri, H. Kwon, S.L. Karstens, Y. Yabuuchi, J. Börgel, J.W. Taylor, K.R. Meihaus, K.C. Bustillo, A.M. Minor, K.A. Persson, C.M. Brown, R.D. Britt, N.P. Stadie, J.R. Long*, *Science*, 390, 808–812 (2025).

39. A Foundation Model for Atomistic Materials Chemistry
I. Batatia[†], P. Benner[†], Y. Chiang[†], A.M. Elena[†], D.P. Kovács[†], J. Riebesell[†], X.R. Advincula, M. Asta, M. Avaylon, W.J. Baldwin, F. Berger, N. Bernstein, A. Bhowmik, F. Bigi, S.M. Blau, V. Cărare, M. Ceriotti, S. Chong, J.P. Darby, S. De, F. Della Pia, V.L. Deringer, R. Elijošius, Z. El-Machachi, E. Fako, F. Falcioni, A.C. Ferrari, J.L.A. Gardner, M.J. Gawkowski, A. Genreith-Schriever, J. George, R.E.A. Goodall, J. Grandel, C.P. Grey, P. Grigorev, S. Han, W. Handley, H.H. Heenen, K. Hermansson, C.H. Ho, S. Hofmann, C. Holm, J. Jaafar, K.S. Jakob, H. Jung, V. Kapil, A.D. Kaplan, N. Karimitari, J.R. Kermode, P. Kourtis, N. Kroupa, J. Kullgren, M.C. Kuner, D. Kuryla, G. Liepuoniute, C. Lin, J.T. Margraf, I.-B. Magdău, A. Michaelides, J.H. Moore, A.A. Naik, S.P. Niblett, S.W. Norwood, N. O'Neill, C. Ortner, K.A. Persson, K. Reuter, **A.S. Rosen**, L.A.M. Rosset, L.L. Schaaf, C. Schran, B.X. Shi, E. Sivonxay, T.K. Stenczel, C. Sutton, V. Svahn, T.D. Swinburne, J. Tilly, C. van der Oord, S. Vargas, E. Varga-Umbrich, T. Vegge, M. Vondrák, Y. Wang, W.C. Witt, T. Wolf, F. Zills, G. Csányi*, *Journal of Chemical Physics*, 163, 184110 (2025).

38. An Accurate and Efficient Framework for Modelling the Surface Chemistry of Ionic Materials
B.X. Shi, **A.S. Rosen**, T. Schäfer, A. Grüneis, V. Kapil, A. Zen, A. Michaelides*, *Nature Chemistry*, 17, 1688–1695 (2025).

37. Computational Investigation of the Impact of Metal–Organic Framework Topology on Hydrogen Storage Capacity
K. Liu, H. Chen, T. Islamoglu, **A.S. Rosen**, X. Wang, O.K. Farha, R.Q. Snurr*, *Molecular Systems Design & Engineering*, 10, 817–835 (2025).

36. Accelerated Data-Driven Materials Science with the Materials Project
M.K. Horton, P. Huck, R.X. Yang, J.M. Munro, S. Dwaraknath, A.M. Ganose, R.S. Kingsbury, M. Wen, J.-X. Shen, T.S. Mathis, A.D. Kaplan, K. Berket, J. Riebesell, J. George, **A.S. Rosen**, E.W.C. Spotte-Smith, M.J. McDermott, O.A. Cohen, A. Dunn, M.C. Kuner, G.-M. Rignanese, G. Hautier, G. Petretto, D. Waroquiers, S.M. Griffin, J.B. Neaton, D.C. Chrzan, M. Asta, S. Cholia, G. Ceder, S.P. Ong, A. Jain, K.A. Persson*, *Nature Materials*, 24, 1522–1532 (2025).

35. Atomate2: Modular Workflows for Materials Science
A.M. Ganose*, H. Sahasrabuddhe, M. Asta, K. Beck, T. Biswas, A. Bonkowski, J. Bustamante, X. Chen, Y.

Chiang, J. Clary, D. Chrzan, O. Cohen, C. Ertural, M. Gallant, J. George, S. Gerits, R. Goodall, R. Guha, G. Hautier, M. Horton, A.D. Kaplan, R.S. Kingsbury, M.C. Kuner, B. Li, X. Linn, M.J. McDermott, R.S. Mohanakrishnan, A.A. Naik, J.B. Neaton, K.A. Persson, G. Petretto, T.A.R. Purcell, F. Ricci, B. Rich, J. Riebesell, G.-M. Rignanese, **A.S. Rosen**, M. Scheffler, J. Schmidt, J.-X. Shen, A. Sobolev, R. Sundararaman, C. Tezak, V. Trinquet, J.B. Varley, D. Vigil-Fowler, D. Wang, D. Waroquiers, M. Wen, H. Yang, H. Zheng, J. Zheng, Z. Zhu, A. Jain*, *Digital Discovery*, 4, 1944–1973 (2025).

34. Computational Modeling of a High-Entropy Alloy for Enhanced Ammonia Synthesis
J.H. Baratta, **A.S. Rosen***, *Chem Catalysis*, 5, 101426 (2025).

33. Machine Learned Potential for High-Throughput Phonon Calculations of Metal-Organic Frameworks
A.M. Elena†, P.D. Kamath†, T.J. Inizan, **A.S. Rosen**, F. Zanca, K.A. Persson*, *npj Computational Materials*, 11, 125 (2025).

Before Princeton

32. A Recipe for Charge Density Prediction
X. Fu*, **A.S. Rosen**, K. Bystrom, R. Wang, A. Musaelian, B. Kozinsky, T. Smidt, T.S. Jaakkola, *Advances in Neural Information Processing Systems (NeurIPS)*, 37, 9727–9752 (2024).

31. Analytical *Ab Initio* Hessian from a Deep Learning Potential for Transition State Optimization
E.C.-Y. Yuan†, A. Kumar†, X. Guan, E.D. Hermes, **A.S. Rosen**, J. Zádor, T. Head-Gordon*, S.M. Blau*, *Nature Communications,* 15, 8865 (2024).

30. Investigating the Behavior of Diffusion Models for Accelerating Electronic Structure Calculations
D. Rothchild, **A.S. Rosen**, E. Taw, C. Robinson, J.E. Gonzalez*, A.S. Krishnapriyan*, *Chemical Science*, 15, 13506–13522 (2024).

29. cclib 2.0: An Updated Architecture for Interoperable Computational Chemistry
E. Berquist†, A. Dumi†, S. Upadhyay†, O.D. Abarbanel, M. Cho, S. Gaur, V. Hugo Cano Gil, G.R. Hutchinson, O.S. Lee, **A.S. Rosen**, S. Schamnad, F.S.S. Schneider, C. Steinmann, M. Stolyarchuk, J.E. Vanderzande, W. Zak, K.M. Langner*, *Journal of Chemical Physics*, 161, 042501 (2024).

28. MOFDiff: Coarse-Grained Diffusion for Metal–Organic Framework Design
X. Fu*, T. Xie, **A.S. Rosen**, T.S. Jaakkola, J.A. Smith*, *International Conference on Learning Representations (ICLR)*, 12 (2024).

27. Structured Information Extraction from Scientific Text with Large Language Models
J. Dagdelen, A. Dunn, S. Lee, N. Walker, **A.S. Rosen**, G. Ceder, K.A. Persson, A. Jain*, *Nature Communications,* 15, 1418 (2024).

26. Jobflow: Computational Workflows Made Simple
A.S. Rosen, M. Gallant, J. George, J. Riebesell, H. Sahasrabuddhe, J.-X. Shen, M. Wen, M.L. Evans, G. Petretto, D. Waroquiers, G.-M. Rignanese, K.A. Persson, A. Jain, A.M. Ganose*, *Journal of Open Source Software,* 9, 5995 (2024).

25. Tetrazine-Linked Covalent Organic Frameworks with Acid Sensing and Photocatalytic Activity
A. Zadehnazari, A. Khosropour, A.A. Altaf, **A.S. Rosen**, A. Abbaspourrad*, *Advanced Materials*, 36, 2311042 (2024).

24. Evidence of a Uranium-Paddlewheel Node in a Catecholate-Based Metal–Organic Framework
J.G. Knapp, X. Wang, **A.S. Rosen**, X. Wang, X. Gong, M. Schneider, T. Elkin, K.O. Kirlikovali, M. Fairley, M.D. Krzyaniak, M.R. Wasielewski, N.C. Gianneschi, R.Q. Snurr, O.K. Farha*, *Angewandte Chemie International Edition*, 62, e202305526 (2023).

23. An Ecosystem for Digital Reticular Chemistry
K.M. Jablonka, **A.S. Rosen**, A.S. Krishnapriyan, B. Smit*, *ACS Central Science*, 9, 563–581 (2023).
— Highlighted in *Chemistry World*

22. Free-Atom-Like *d* States Beyond the Dilute Limit of Single-Atom Alloys
A.S. Rosen, S. Vijay, K.A. Persson*, *Chemical Science*, 14, 1503–1511 (2023).

21. Effect of Composition and Local Environment on CO_2 Adsorption on Nickel and Magnesium Oxide Solid Solutions
A. Peng, **A.S. Rosen**, R.Q. Snurr*, H. Kung*, *Journal of Physical Chemistry C*, 126, 19705–19714 (2022).

20. A Flexible and Scalable Scheme for Mixing Computed Formation Energies from Different Levels of Theory
R. Kingsbury, **A.S. Rosen**, A. Gupta, J. Munro, S.P. Ong, A. Jain, S. Dwaraknath, M.K. Horton, K.A. Persson*, *npj Computational Materials*, 8, 195 (2022).

19. High-Throughput Predictions of Metal–Organic Framework Electronic Properties: Theoretical Challenges, Graph Neural Networks, and Data Exploration
A.S. Rosen*, V. Fung, P. Huck, C.T. O'Donnell, M.K. Horton, D.G. Truhlar, K.A. Persson, J.M. Notestein, R.Q. Snurr, *npj Computational Materials*, 8, 112 (2022).
— Highlighted in Nature Portfolio's Nobel Prize in Physics 2024 collection
— Highlighted in *MRS Bulletin*, 47, 886 (2022).

18. Exploring Mechanistic Routes for Light Alkane Oxidation with an Iron-Triazolate Metal–Organic Framework
A.S. Rosen, J.M. Notestein*, R.Q. Snurr*, *Physical Chemistry Chemical Physics*, 24, 8129–8141 (2022).
— Selected by the editors as a 2022 HOT *PCCP* article.

17. Realizing the Data-Driven, Computational Discovery of Metal–Organic Framework Catalysts
A.S. Rosen, J.M. Notestein*, R.Q. Snurr*, *Current Opinion in Chemical Engineering*, 35, 100760 (2022).

16. Fine-Tuning a Robust Metal–Organic Framework Towards Enhanced Clean Energy Gas Storage
Z. Chen, M.R. Mian, S.-J. Lee, H. Chen, X. Zhang, K.O. Kirlikovali, S. Shulda, P. Melix, **A.S. Rosen**, P.A. Parilla, T. Gennett, R.Q. Snurr, T. Islamoglu*, T. Yildirim*, O.K. Farha*, *Journal of the American Chemical Society*, 143, 18838–18843 (2021).

15. Machine Learning the Quantum-Chemical Properties of Metal–Organic Frameworks for Accelerated Materials Discovery
A.S. Rosen*, S.M. Iyer[u], D. Ray, Z. Yao, A. Aspuru-Guzik, L. Gagliardi, J.M. Notestein, R.Q. Snurr, *Matter*, 4, 1578–1597 (2021).
— Featured on the cover of *Matter* and previewed in *Patterns*, 2, 100239 (2021).

14. Zr_6O_8 Node-Catalyzed Butene Hydrogenation and Isomerization in the Metal–Organic Framework NU-1000
K.E. Hicks, **A.S. Rosen**, Z.H. Syed, R.Q. Snurr, O.K. Farha*, J.M. Notestein*, *ACS Catalysis*, 10, 14959–14970 (2020).

13. Supramolecular Porous Assemblies of Atomically Precise Catalytically Active Cerium-Based Clusters
M.C. Wasson, X. Zhang, K. Otake, **A.S. Rosen**, S. Alayoglu, M.D. Krzyaniak, Z. Chen, L.R. Redfern, L. Robison, F.A. Son, Y. Chen, T. Islamoglu, J.M. Notestein, R.Q. Snurr, M.R. Wasielewski, O.K. Farha*, *Chemistry of Materials*, 32, 8522–8529 (2020).

12. Comparing GGA, GGA+*U*, and Meta-GGA Functionals for Redox-Dependent Binding at Open Metal Sites in Metal–Organic Frameworks
A.S. Rosen, J.M. Notestein*, R.Q. Snurr*, *Journal of Chemical Physics*, 152, 224101 (2020).

11. Topological Effects on Separation of Alkane Isomers in Metal–Organic Frameworks
N.S. Bobbitt, **A.S. Rosen**, R.Q. Snurr*, *Fluid Phase Equilibria*, 519, 112642 (2020).

10. High-Valent Metal-Oxo Species at the Nodes of Metal–Triazolate Frameworks: The Effects of Ligand Exchange and Two-State Reactivity for C–H Bond Activation

A.S. Rosen, J.M. Notestein*, R.Q. Snurr*, *Angewandte Chemie International Edition*, 59, 19494–19502 (2020).

9. Tuning the Redox Activity of Metal–Organic Frameworks for Enhanced, Selective O_2 Binding: Design Rules and Ambient Temperature O_2 Chemisorption in a Cobalt–Triazolate Framework
   A.S. Rosen, M.R. Mian, T. Islamoglu, H. Chen, O.K. Farha, J.M. Notestein, R.Q. Snurr*, *Journal of the American Chemical Society*, 142, 4317–4328 (2020).

8. Identification Schemes for Metal–Organic Frameworks to Enable Rapid Search and Cheminformatics Analysis
   B.J. Bucior, **A.S. Rosen**, M. Haranczyk, Z. Yao, M.E. Ziebel, O.K. Farha, J.T. Hupp, J.I. Siepmann, A. Aspuru-Guzik, R.Q. Snurr*, *Crystal Growth and Design*, 19, 6682–6697 (2019).

7. Structure–Activity Relationships that Identify Metal–Organic Framework Catalysts for Methane Activation
   A.S. Rosen, J.M. Notestein*, R.Q. Snurr*, *ACS Catalysis*, 9, 3576–3587 (2019).
   — Featured in C&EN

6. Identifying Promising Metal–Organic Frameworks for Heterogeneous Catalysis via High-Throughput Periodic Density Functional Theory
   A.S. Rosen, J.M. Notestein*, R.Q. Snurr*, *Journal of Computational Chemistry*, 40, 1305–1318 (2019).

5. Evidence for Copper Dimers in Low-Loaded CuO_x/SiO_2 Catalysts for Cyclohexane Oxidative Dehydrogenation
   S.L. Nauert, **A.S. Rosen**, H. Kim, R.Q. Snurr, P.C. Stair, J.M. Notestein*, *ACS Catalysis*, 8, 9775–9789 (2018).

4. Comprehensive Phase Diagrams of MoS_2 Edge Sites Using Dispersion-Corrected DFT Free Energy Calculations
   A.S. Rosen, J.M. Notestein*, R.Q. Snurr*, *Journal of Physical Chemistry C*, 122, 15318–15329 (2018).

3. Correlations, Trends and Potential Biases among Publicly Accessible Web-Based Student Evaluations of Teaching: A Large-Scale Study of RateMyProfessors.com Data
   A.S. Rosen*, *Assessment and Evaluation in Higher Education*, 43, 31–44 (2018).
   — Featured in *Inside Higher Ed* and VOA News

2. A Detailed Combined Experimental and Theoretical Study on Dimethyl Ether/Propane Blended Oxidation
   E.E. Dames, **A.S. Rosen**, B.W. Weber, C.W. Gao, C.-J. Sung, W.H. Green*, *Combustion and Flame*, 168, 310–330 (2016).

1. Validation of X-ray Line Ratios for Electron Temperature Determination in Tokamak Plasmas
   A.S. Rosen, M.L. Reinke*, J.E. Rice, A.E. Hubbard, J.W. Hughes, *Journal of Physics B*, 47, 105701 (2014).
   — Selected as an article representing "the best work published in the *Journal of Physics B* in 2014"

Teaching

CBE 504: Chemical Kinetics and Reaction Engineering Fall 2026
Enrollment: 20

CBE 423/MSE 423: Computational Materials Design and Discovery Spring 2026
Overall Course Quality: **4.82/5.0**, Instructor Lecture Quality: **4.82/5.0**, Enrollment: 29

- Selected quotes: "Professor Rosen is a talented lecturer and is a master at breaking down complex topics to a level where everyone can understand", "I would definitely recommend this course to ANYONE remotely interested!", "I was also super happy to discover that Prof. Rosen is the most passionate professor I had in my 4 years here."
- Received Commendation for Outstanding Teaching

CBE 504: Chemical Kinetics and Reaction Engineering Fall 2025

Overall Course Quality: **4.82/5.0**, Instructor Lecture Quality: **4.88/5.0**, Enrollment: 24

- Selected quotes: "Andrew Rosen is a gifted instructor. His lectures are well thought out, easy to follow, and engaging", "Andrew was probably the best professor I had this semester, and never failed to correlate the material in lecture to things that we might actually be interested in."
- Received Commendation for Outstanding Teaching

CBE 504: Chemical Reactor Engineering Fall 2024

Overall Course Quality: **4.77/5.0**, Instructor Lecture Quality: **4.90/5.0**, Enrollment: 33

- Selected quotes: "Lectures were stunning", "Dr. Andrew Rosen is hands-down one of the best instructors", "Prof. Rosen basically wrote a textbook for us… it's 6.02x10^23 better than every kinetics textbook I used in undergrad"
- Received Commendation for Outstanding Teaching

Research Grants

7. **Breakthrough Energy Foundation Explorer Fellow Grant**
   AI-Guided Discovery of Heterogeneous Catalysts with Unbound Potential
   Single PI. Award: confidential per sponsor agreement (11/2025–10/2026).

6. **NSF Cyberinfrastructure for Sustained Scientific Innovation**, Office of Advanced Cyberinfrastructure
   Collaborative Research: Frameworks: SINAPSE: Scalable Infrastructure for AI-coupled Predictive Simulation Enhancement
   Co-PI. Award to Rosen: $725,000 (10/2025–09/2028).

5. **New Industrial Collaborations Grant**, Meta and Princeton Office of the Dean for Research
   Development of an AI-Ready Dataset for Next-Generation Battery Materials
   Single PI. Award: $275,000 (07/2025–06/2027).

4. **SEAS Innovation Grant**, Princeton SEAS
   A Combined Experimental and Computational Approach for Accelerated Zeolite Discovery via Pre-Nucleation Building Units
   Lead PI. Award: $230,000. Award to Rosen: $100,000 (01/2026–12/2027).

3. **SEAS Innovation Seed Grant**, Princeton SEAS
   Democratizing the Computational Discovery of Clean Energy Materials
   Single PI. Award: $50,000 (06/2025–08/2025).

2. **NSF Materials Research Science and Engineering Center Seed Grant**, Princeton Center for Complex Materials
   Developing Thermodynamic and Kinetic Stability Relationships for Porous Framework Materials
   Single PI. Award: $60,000 (02/2025–01/2026).

1. **AI Lab Seed Grant**, Princeton Laboratory for Artificial Intelligence
   AI-Accelerated Discovery of Porous Materials with Confined Electrons for Challenging Chemical Transformations
   Lead PI. Award: $90,000. Award to Rosen: $45,000 (01/2025–06/2026).

Computing Grants

6. **NERSC AI for Science Award**: "Machine-Learned Interatomic Potentials to Investigate the Thermal and Mechanical Stability of Metal–Organic Frameworks." Single PI. Award: 8000 GPU node hours (06/2026–01/2027).

5. **ALCF Director's Discretionary Award**: "An Off-Equilibrium Metal-Organic Framework Dataset for Training Machine-Learned Potentials." Single PI. Award: 1000 TB valued at $50,000 (09/2025–10/2027).

4. **NERSC High-Impact Science at Scale Award**: "Assessing the Thermodynamic Stability Limits of Nanoporous Materials." Single PI. Award: 25,000 GPU node hours, 5000 CPU node hours (06/2025–12/2025).

3. **NERSC AI for Science Award**: "Development of an Interpretable Machine-Learned Interatomic Potential for Chemical Bonding in Energy Materials." Single PI. Award: 8000 GPU node hours (06/2025–01/2026).

2. **NSF National Artificial Intelligence Research Resource Pilot**: "Building a Foundational Metal–Organic Framework Dataset for Machine Learning Potentials." Single PI. Award: 100,000 GPU hours valued at $164,000 (04/2025–03/2026).

1. **NSF ACCESS**: "Developing Thermodynamic and Kinetic Stability Relationships for Porous Materials." Single PI. Award: 200,000 ACCESS Credits (01/2025–01/2026).

Research Mentoring

Current Mentees

Postdoctoral Researchers

- Dr. Sihoon Choi, Gordon & Betty Moore Foundation Postdoctoral Fellow (2026 – present), DataX Postdoctoral Research Associate (2025 – 2026)
- Dr. Yuliang Shi, Postdoctoral Research Associate (2026 – present)
- Dr. Hananeh Oliaei, Postdoctoral Research Associate (2025 – present)

Graduate Students

- Jaehong ("Russel") Kwon, CBE, co-advised with Prof. David Graves (2026 – present)
- Isabella ("Izzie") Furrick, CBE (2026 – present)
- Julia Baratta, CBE (2025 – present)
- Blake Dallmann, CBE (2025 – present)
- Naisargi Goyal, CBE (2025 – present)

Undergraduate Researchers (* = senior thesis)

- Jiacheng ("Anthony") Shen, CBE 351, CBE Class of '28 (2026 – present)
- Aryan Saha*, CBE 351, ECE 298/299/398/399/498, OURSIP ×2, ECE Class of '27 (2024 – present)

University Service

University Committees, Princeton University

- *Member*, Committee on Examinations and Standing (2025 – present)

Department of Chemical and Biological Engineering Committees, Princeton University

- *Member*, Undergraduate Committee (2024 – present)
- *Advisor*, CBE Class of '28 (2025 – present)
- *Faculty Representative*, Diversity, Equity, Climate, and Inclusion Committee (2026 – present)
- *Pre-major advisor*, CBE rising sophomores (2024 – 2026)
- *Member*, IT Support Specialist Hiring Committee (2026)
- *Member*, Graduate Committee (2024 – 2025)

Graduate Student Thesis and First Proposition Committees, Princeton University

- 18× CBE students, 1× CEE student, 1× GEO student

Undergraduate Academic Advising, Princeton University

- CBE academic advisor, 13× CBE students
- Pre-major advisor, 16× BSE students

Senior Thesis Second Reader and Evaluator, Princeton University

- Second Reader for 4× CBE students
- Poster Evaluator for 7× CBE students

Princeton Research Computing
- Stellar AI Steering Committee (2026 – present)

AI Lab, Princeton University
- Faculty advisor for the Machine Learning in Chemical Physics journal club (2024 – present)
- Reviewer, AI Lab Seed Grant (2025)

Center for Statistics and Machine Learning, Princeton University
- Program evaluator, Statistics and Machine Learning Minor (2025)

Princeton Plasma Physics Laboratory joint with Princeton University
- Reviewer, Collaborative Research in Plasma Science and Technology Grant (2026)

Nassau Hall Society, Princeton University
- Panelist, "AI Solving the Unsolvable" (2025)

External Service

Manuscript Peer-Review

ResearcherID: G-2407-2014
- 90+ reviews across Nature, Nature Machine Intelligence, Nature Reviews Chemistry, Nature Communications, npj Computational Materials, Science Advances, Journal of the American Chemical Society, JACS Au, Angewandte Chemie, Advanced Functional Materials, ACS Materials Letters, ACS Applied Materials and Interfaces, Journal of Chemical Information and Modeling, Journal of Physical Chemistry C, Matter, Cell Reports Physical Science, Digital Discovery, Electronic Structure, Scientific Data, PLoS One, Journal of Open Source Software, F1000Research, STAM Methods, Assessment and Evaluation in Higher Education

Grant and Fellowship Proposal Peer-Review
- Department of Energy Office of Science Graduate Student Research Program, Basic Energy Sciences (2026)
- NSF Division of Materials Research (2026)
- Air Force Office of Scientific Research, Department of Defense (2026)
- Department of Energy, Basic Energy Sciences (2026)
- NSF Division of Chemical, Bioengineering, Environmental and Transport Systems (2025)
- Petroleum Research Fund, American Chemical Society (2025)
- Swiss National Supercomputing Centre (2025)
- Department of Energy, Basic Energy Sciences (2024)
- Petroleum Research Fund, American Chemical Society (2023)
- Swiss National Supercomputing Centre (2023)

Atomic Simulation Environment

Steering Committee (2025 – present)
- Guide the future directions of the Atomic Simulation Environment, which receives >2M downloads per month

Materials Project Software Foundation

Board member and co-founder (2023 – 2026)
- Guide the future directions of the Materials Project software ecosystem, which has >600k registered users

Open-Source Software Development
- Creator and maintainer of the widely adopted QMOF Database, which has been downloaded >14k times
- Maintainer and lead developer of the Quantum Accelerator (QuAcc), which receives >9k downloads per month and is adopted in industry (e.g. by Meta)
- Co-developer of popular atomistic simulation packages, including the Atomic Simulation Environment (>2M downloads per month), Pymatgen (>820k downloads per month), and Atomate2 (>24k downloads per month)
- Contributed to over 100 packages in computational chemistry, materials science, and workflow orchestration

Conference Activity

- Co-organizer for the CECAM Flagship Workshop on "Physics-Aware Machine Learning for Molecules and Materials" (2026)
- Chair for "New Developments in Computational Catalysis III: Advances in Fundamental Understanding of Catalysis," Chemical Reaction Engineering Division, AIChE Annual Meeting (2026)
- Selection Committee for the AIChE Inorganic Materials Graduate Student Award, Materials Engineering and Sciences Division, AIChE Annual Meeting (2026)
- Bin co-leader for "New Developments in Computational Catalysis," Chemical Reaction Engineering Division, AIChE Annual Meeting (2026)
- Co-chair for "Special Session in Honor of Prof. Randall Snurr's 60th Birthday," Computational Molecular Science & Engineering Forum, AIChE Annual Meeting (2026)
- Co-chair for "Accelerated Discovery of Inorganic Materials: High-Throughput Experiments, Modeling, and Data Science," Materials Engineering and Sciences Division, AIChE Annual Meeting (2025)
- Poster judge, Materials Engineering and Sciences Division, AIChE Annual Meeting (2023, 2025)
- Poster judge, Chemical Reaction Engineering Division, AIChE Annual Meeting (2023)
- Co-chair for "Data Science for Catalysis," ACS Fall Meeting (2023)

AIChE Chemical Reaction Engineering Division

Social Media Team (2019 – 2024)

- Organize social media campaigns dedicated to highlighting underrepresented and early-career chemical engineers

Letters to a Pre-Scientist

Scientist PenPal (2019 – 2024)

- Outreach to establish relationships between scientists and middle school students in low-income classrooms

Rosen Review

Chemical Engineering Review Website (2011 – 2024)

- Created an educational website viewed 1,000,000+ times with visitors from 191 countries

Strategic Engagements and Science Policy

2. "AI × Material Innovation: Breakthroughs for a Clean Energy Future Workshop." **Bezos Earth Fund**, 2025.

1. "Semiconductor Materials Accelerator Open Roundtable." **White House Office of Science and Technology Policy**, 2024.

Invited Presentations

Forthcoming Presentations

38. **APS Global Physics Summit 2027**, Machine Learning for Atomistic Simulation, Atlanta, GA, 2027.

37. **ACS Spring 2027 Meeting**, CATL: Bridging Adsorbate and Gas-Phase Thermochemistry, New Orleans, LA, 2027.

36. **ACS Spring 2027 Meeting**, COMP: Frontiers in AI and Molecular Modeling, New Orleans, LA, 2027.

35. **5th Texas Pore Engineering Conference (TEXPEC)**, UT Austin, 2026.

34. **2026 AIChE Annual Meeting**, CoMSEF, Minneapolis, MN, 2026.

33. **Rowan University**, Glassboro, NJ, 2026.

32. **Meta's Open Science Summit 2026**, Menlo Park, CA, 2026.

Prior Presentations

31. "Data-Driven Design of Materials with Exotic Electronic Character." **NSF Institute for Data-Driven Dynamical Design**, Princeton University, 2026.

30. "Combining Machine Learning and Quantum Chemistry to Guide Materials Design." **Princeton University**, AI

for Accelerating Invention and the Center for Statistics and Machine Learning, 2025.

29. "The Quantum Accelerator: Accessible and Scalable Materials Science Workflows." **CECAM**, The Atomic Simulation Environment Ecosystem: Present and Perspectives, 2025.

28. "Quantum-Chemical Engineering and Computational Materials Design." **Princeton University**, Princeton Catalysis Initiative Symposium, 2025.

27. "Discovering Unique Electronic Structure Properties in Solid-State Materials via High-Throughput Simulations." **Princeton University**, Princeton Materials Institute Symposium, 2025.

26. "Discovering Unique Electronic Structure Properties in Solid-State Materials via High-Throughput Simulations." **Princeton Plasma Physics Laboratory**, Computational Sciences Department, 2025.

25. "Designing Heterogeneous Catalysts with Free-Atom-Like Electronic States." **Catalysis Society of Metropolitan New York**, Lehigh University, 2025.

24. "Discovering Unique Electronic Structure Properties in Solid-State Materials via High-Throughput Simulations." **New Jersey Institute of Technology,** Department of Chemistry and Environmental Science, 2025.

23. "Guiding the Design of New Materials with Electronic Structure Calculations in the Big Data Regime." **Princeton University**, Chemistry in Solutions and at Interfaces – Computational Chemical Science Center, 2024.

22. "A Data-Driven Approach to Tailor the Electronic Structure Properties of Materials." **Drexel University**, Department of Chemistry, 2024.

21. "Discovering Unique Electronic Structure Properties in Solid-State Materials via High-Throughput Simulations." **Rutgers University–Newark**, Department of Physics, 2024.

20. "Accessible and Interoperable Computational Workflows to Satisfy the Data-Hungry Machines." **AI for Multidisciplinary Exploration and Discovery Workshop on Heterogeneous Catalysis**, Chicago, IL, 2024.

19. "Combining High-Throughput Workflows, Quantum Chemistry, and AI for the Discovery of Tunable Materials with Unprecedented Properties." **University of Pennsylvania**, Penn Institute for Computational Science, 2024.

18. "The Quantum Accelerator: Accessible and Scalable Materials Science Workflows." **ParslFest**, 2024.

17. "Discovering Heterogeneous Catalysts with Unique Electronic Structure Properties." **Flatiron Institute,** Initiative for Computational Catalysis Inaugural Workshop, 2024.

Before Princeton

16. "Free-Atom-Like *d*-States Beyond the Dilute Limit of Single-Atom Alloys." **ACS Fall Meeting**, 2023.

15. "The Present and Future of Materials Databases: Metal–Organic Framework Edition." **Materials Research Data Alliance Annual Meeting**, 2023.

14. "Discovering Tunable Materials with Unprecedented Properties via High-Throughput Quantum Chemistry." **Massachusetts Institute of Technology**, Department of Materials Science and Engineering, 2023.

13. "Discovering Tunable Materials with Unprecedented Properties via High-Throughput Quantum Chemistry." **University of California, San Diego**, Department of NanoEngineering, 2023.

12. "Discovering Tunable Materials with Unprecedented Properties via High-Throughput Quantum Chemistry." **University of Colorado Boulder**, Department of Chemical and Biological Engineering, 2023.

11. "Discovering Tunable Materials with Unprecedented Properties via High-Throughput Quantum Chemistry." **University of Illinois, Urbana-Champaign**, Department of Materials Science and Engineering, 2023.

10. "Discovering Tunable Materials with Unprecedented Properties via High-Throughput Quantum Chemistry." **Yale University**, Department of Chemical and Environmental Engineering, 2023.

9. "Discovering Tunable Materials with Unprecedented Properties via High-Throughput Quantum Chemistry." **University of California, Santa Barbara**, Department of Chemical Engineering, 2023.

8. "Discovering Tunable Materials with Unprecedented Properties via High-Throughput Quantum Chemistry."

University of Washington, Department of Chemical Engineering, 2023.

7. "Discovering Tunable Materials with Unprecedented Properties via High-Throughput Quantum Chemistry." **Princeton University**, Department of Chemical and Biological Engineering, 2023.
6. "Navigating the Metal–Organic Framework Universe with High-Throughput Quantum Chemistry." **Statistical Thermodynamics and Molecular Simulations Virtual Seminar Series**, 2022.
5. "Discovering Tunable Materials with Unprecedented Properties Using High-Throughput Quantum Chemistry." **Distinguished Young Scholars Seminar Series**, University of Washington, 2022.
4. "The QMOF Database: Accelerating the Discovery of Metal–Organic Frameworks with Targeted Electronic Structure Properties." **Open Databases Integration for Materials Design Workshop**, 2021.
3. "A Guided Journey Through the Metal–Organic Framework Universe: New Materials for Longstanding Challenges." **Northwestern University**, Distinguished Graduate Researcher Award Presentation, 2020.
2. "High-Valent Metal-Oxo Species for C–H Activation and Where to Find Them: A Computationally Guided Expedition." **Catalysis Club of Chicago Symposium**, 2020.
1. "Combining Quantum Chemistry and Supercomputing to Accelerate the Discovery of Promising Metal–Organic Frameworks." **Northwestern SPIE-MRSEC Seminar Series**, 2020.